

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**

| Section / Topic                     | Focus Area                                                | No. of Questions | Marks |
|-------------------------------------|-----------------------------------------------------------|------------------|-------|
| Scatterplots & Bubble Charts        | Scatterplots and bubble charts                            | 2                | 2     |
| Correlation Analysis                | Correlogram and feature selection                         | 2                | 2     |
| Dimensionality Reduction Techniques | PCA, t-SNE, UMAP                                          | 3                | 3     |
| Density & Overplotting Techniques   | Hexbin plots and overplotting handling                    | 2                | 2     |
| Network Visualization               | Nodes and edges in network graphs                         | 2                | 2     |
| Time Series Analysis                | Moving average, detrending, smoothing, seasonality, noise | 5                | 5     |

|                          |                                                                      |              |   |
|--------------------------|----------------------------------------------------------------------|--------------|---|
| Geospatial Visualization | Geospatial plots, map projections, choropleth maps, spatial heatmaps | 4            | 4 |
| <b>GRAND TOTAL:</b>      |                                                                      | <b>Marks</b> |   |

## ASSESSMENT TOOL 2 – MOCK INTERVIEW (20 Questions)

|                         |                                                                                                           |                      |                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------------|----------------------|------------------------------------|
| <b>Course Code:</b>     | DSA06                                                                                                     | <b>Course Title:</b> | Data Handling and Visualization    |
| <b>CO Assessed:</b>     | CO4 – Naturalization(BL5)                                                                                 | <b>Assessment:</b>   | Assessment Tool 2 – Mock Interview |
| <b>Weightage:</b>       | 20% of CIA                                                                                                | <b>Total Marks:</b>  | 20 (20 Questions × 1 Mark)         |
| <b>CO4 Statement:</b>   | <i>Evaluate visualization designs based on perception, cognition, and effectiveness principles (BL5).</i> |                      |                                    |
| <b>Student Name:</b>    | v.surya satvika                                                                                           | <b>Reg. No.:</b>     | 192524254                          |
| <b>Section / Batch:</b> | 2025/d                                                                                                    | <b>Date:</b>         |                                    |

### Code of Conduct

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_

### Instructions

- This test contains 20 questions, each carrying 1 mark. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

### Annexure A – Mock Interview

#### Section: Mock Interview – Answers

#### 1. What is the purpose of a histogram?

A histogram is used to show the frequency distribution of numerical data by grouping values into intervals or bins.

#### 2. How does a box plot help identify outliers?

A box plot identifies outliers as data points lying below  $Q1 - 1.5 \times IQR$  or above  $Q3 + 1.5 \times IQR$ .

**3. What information does a violin plot provide?**

A violin plot shows the distribution, spread, density, and central tendency of numerical data.

**4. What is the difference between a bar chart and a histogram?**

A bar chart compares categorical data, while a histogram represents the distribution of continuous numerical data.

**5. Why are heatmaps useful in data analysis?**

Heatmaps make patterns, relationships, and variations in data easy to identify using colors.

**6. What does color intensity represent in a heatmap?**

Color intensity represents the magnitude or strength of a value; darker or stronger colors usually indicate higher values.

**7. What is a pair plot used for?**

A pair plot is used to visualize relationships and correlations between multiple numerical variables.

**8. How can a density plot help understand data distribution?**

A density plot shows the shape, concentration, peaks, and spread of data using a smooth curve.

**9. What is meant by dimensionality reduction?**

Dimensionality reduction is the process of reducing the number of features or dimensions while retaining important information.

**10. Why is PCA commonly used before visualization?**

PCA reduces high-dimensional data to two or three principal components, making it easier to visualize.

**11. What is a principal component?**

A principal component is a new variable formed as a linear combination of the original variables, capturing maximum variance in the data.

**12. How does UMAP preserve data structure?**

UMAP preserves important local and global relationships between data points while reducing dimensions.

**13. What is the main goal of t-SNE visualization?**

The main goal of t-SNE is to represent high-dimensional data in a low-dimensional space while preserving local similarities.

**14. What does a contour plot represent?**

A contour plot represents three-dimensional data on a two-dimensional graph using lines of equal value.

**15. How are contour lines interpreted in a contour plot?**

Each contour line connects points having the same value. Closely spaced lines indicate rapid changes.

**16. What is a radar chart used for?**

A radar chart is used to compare multiple variables or characteristics across different categories or objects.

**17. What does each axis in a radar chart represent?**

Each axis represents a different variable or performance measure.

**18. How does a parallel coordinates plot visualize multivariate data?**

It represents each variable as a parallel axis, with each observation shown as a line connecting its values across the axes.

**19. What is overplotting in data visualization?**

Overplotting occurs when too many data points overlap, making individual points and patterns difficult to see.

**20. Name one visualization technique used for high-dimensional data.**

PCA (Principal Component Analysis) is one visualization technique used for high-dimensional data.

**Rubrics for Calculation**

| Weigh<br>Criteria tage             |     | Level 4 –<br>Exemplary                                                     | Level 3 –<br>Proficient                             | Level 2 –<br>Developing                                           | Level 1 –<br>Beginning                             | Marks |  |
|------------------------------------|-----|----------------------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------|-------|--|
| Understanding<br>of Concepts       | 50% | Demonstrate<br>s thorough<br>and accurate<br>understandin<br>g of concepts | Shows good<br>understanding<br>with minor<br>errors | Basic<br>understandin<br>g with limited<br>clarity                | Poor<br>understanding<br>or incorrect<br>responses |       |  |
| Critical<br>Thinking &<br>Response | 50% | Provides<br>logical,<br>confident,<br>and<br>wellstructured<br>responses   | Gives relevant<br>responses with<br>minor gaps      | Responses<br>are partially<br>correct with<br>limited<br>analysis | Unable to<br>analyse or<br>respond<br>effectively  |       |  |
| Faculty In-charge                  |     |                                                                            | Course Coordinator                                  |                                                                   | Head of Departmen                                  |       |  |

|                  |                  |                  |
|------------------|------------------|------------------|
|                  |                  |                  |
| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: _____      | Date: _____      | Date: _____      |